TIMBRE MORPHING VIA WEIGHT INTERPOLATION OF SPECIALIZED VARIATIONAL AUTOENCODERS

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ABSTRACT

Timbre morphing, the generation of intermediate timbres between known sounds, expands the sonic palette for music production and sound design. Common techniques interpolate in the latent space of a single generative model trained on multiple timbres, which in small models compromises the fidelity of each one. We propose an alternative: training variational autoencoders (β-VAE) specialized per timbre, and generating hybrid timbres by interpolating directly in the model weight space. The system is trained on STFT magnitude spectrograms from four sources (piano, guitar, voice and bass) and evaluated with Fréchet Audio Distance over MERT embeddings. Specializing each model by fine-tuning from a common base model, rather than training it from scratch, yields more gradual and predictable timbre transitions, especially when combined with latent-space regularization (β-VAE). Being built on small, lightweight models, the approach is potentially suitable for real-time synthesis.

1. INTRODUCTION

Timbre is the perceptual attribute that lets us distinguish two sounds of equal pitch and loudness produced by different sources, such as the same note played on a piano and on a guitar. Generating intermediate timbres between two known sounds—commonly called timbre *morphing*—is of particular interest for music production and sound design: it enables exploring hybrid sonorities that belong to no single source and thus expands the palette beyond traditional timbres.

To synthesize such sonorities from familiar ones it is natural to draw on the generative power of deep neural networks. The most widespread approaches learn a single generative model over multiple timbres and interpolate within its latent space; however, when the model is small this strategy tends to compromise the fidelity of each individual timbre. This work explores an alternative: training a specialized model per timbre and interpolating directly in the space of their *weights*, so as to preserve the reconstruction quality of each source model when generating the intermediate states.

A design principle that runs through the whole proposal is to keep the models deliberately small. We want them easy to train and to run, so that both inference and the interpolation of their weights can be carried out with modest hardware and, eventually, in real time. This orientation aims at a practical tool for musical performance and production, and it conditions the architecture and data decisions taken throughout.

Contributions. (i) We propose and validate a timbre-morphing technique based on interpolating the *weights* of per-instrument specialized VAEs, instead of interpolating in the latent space as is usual in the literature. (ii) We provide empirical evidence that two factors are decisive for high-quality interpolations: latent-space regularization (β-VAE) and initializing the models from a common pretrained checkpoint, with regularization being the more impactful. (iii) We show that FAD combined with MERT embeddings is a robust and consistent metric for timbre similarity, outperforming cosine similarity in stability.

2. RELATED WORK

Audio morphing can be organized by *where* the interpolation that yields the intermediate sound takes place: on the **signal**, on the **latent space** of a generative model, or—as we propose—on the model **weights**. A foundational signal-domain method [1] aligns two spectrograms with dynamic time warping and cross-fades them; it is training-free but requires explicit alignment and was validated only on voice. With deep learning, morphing moved to the latent space: NSynth [2] interpolates note embeddings in a *single* WaveNet-decoder autoencoder trained on all instruments, materialized by Magenta as an interactive grid [3] close to the tool motivating our work; its autoregressive decoder, however, precludes real-time synthesis.

Recent work relies on latent diffusion, with high quality at high cost. Latent diffusion bridges [4] train one diffusion model per instrument connected by a shared prior, but perform full timbre *transfer* rather than controllable intermediate states. SoundMorpher [5] morphs in the noise and embedding space of a pretrained text-to-audio model and introduces perceptually uniform morphing, at the cost of a large model. For guitar tones, Chen *et al.* [6] find that a lightweight autoencoder beats fine-tuned diffusion, and one variant interpolates network weights—the closest contact point with our idea, though restricted to a single instrument. Mix2Morph [7] trains a large text-to-audio model to morph from noisy mixes. Overall, the literature interpolates the *content* inside a fixed model; interpolating the *parameterization* of distinct, lightweight per-instrument models is the space this work occupies. We also adopt FAD, on which the community has converged for timbre similarity.

Variational autoencoders (VAE) [8] impose a probabilistic structure on the latent space via the Kullback–Leibler (KL) divergence, which facilitates sampling and interpolation; the β-VAE [9] weights this term by a scalar β that balances reconstruction quality against regularization. For music sound modeling, variational models offer the best fidelity/regularization trade-off [10]; RAVE [11] showed real-time VAE synthesis; and sparse autoencoders build perceptually consistent timbre spaces [12].

3. METHOD

3.1 Dataset

We work with four instruments: piano, guitar, bass and voice. They are familiar and easily recognizable sources with clearly distinguishable timbres, a necessary condition for evaluating timbre transitions. We gathered approximately 87 minutes of real recordings per instrument (about 5.8 hours in total), keeping the amount balanced to avoid biasing training. Piano was collected from public-domain classical recordings; guitar from a subset of GuitarSet [13] (electric guitar, melodic and chordal passages); bass from jazz-standard basslines in FiloBass [14]; and voice from a representative subset of VocalSet [15] (two male and two female voices, with melodies, scales and arpeggios).

3.2 Audio Representation and Preprocessing

All tracks are converted to mono and resampled to 22050 Hz. We then apply the Short-Time Fourier Transform (STFT) with a 2048-sample Hann window and a 512-sample hop, keeping magnitude and phase separately. Since timbral information resides mainly in the spectral energy distribution, the model works only with the magnitude, expressed in decibels as $S = 10 \log_{10}(|F|^2)$, where F is the complex spectrum. We clip values below -60 dB, shift the energy floor to zero, and divide by the global maximum X_{max} (stored with the model) to map magnitudes to $[0, 1]$. Each spectrogram frame is thus a 1025-dimensional vector (*win_length*$/2 + 1$ bins), the unit fed to the network; frames are processed independently, without explicitly modeling their temporal evolution.

3.3 Variational Autoencoder

The VAE is a latent-variable generative model that assumes data are generated from a latent variable z with prior $p(z) = \mathcal{N}(0, I)$. The decoder models the likelihood $p(x \mid z)$ and the encoder the approximate posterior $q(z \mid x)$. Training maximizes the Evidence Lower Bound (ELBO), i.e. minimizes a reconstruction term plus a KL regularizer. The β-VAE [9] weights the KL term by β:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta\, \mathcal{L}_{\text{KL}}, \tag{1}$$

where under a Gaussian likelihood $\mathcal{L}_{\text{rec}}$ reduces to the mean squared error (MSE) between input and reconstruction, and $\mathcal{L}_{\text{KL}}$ is the closed-form KL divergence between $q(z \mid x)$ and $\mathcal{N}(0, I)$. The latent vector is obtained via the reparameterization trick $z = \mu + \varepsilon \odot \exp(\frac{1}{2} \log \sigma^2)$, $\varepsilon \sim \mathcal{N}(0, I)$; when $\beta = 0$ the KL term vanishes, the model reduces to a deterministic autoencoder (AE) and we take $z = \mu$. Higher β pushes the posteriors toward the prior, yielding a smoother, more continuous latent space at the cost of reconstruction fidelity.

3.4 Architecture

The encoder is a stack of fully connected layers that progressively reduce dimensionality from the 1025 input values down to the latent space, following $1025 \to (512, 256, 128, 64, 32, 16, 8) \to 4$; each hidden layer combines a linear transformation, Batch Normalization and an ELU activation. The variational encoder outputs a mean vector μ and a log-variance vector $\log \sigma^2$. The decoder mirrors this structure, expanding z back to a 1025-dimensional magnitude spectrum; its final layer is linear (no activation) so as not to restrict the reconstructed range. We use a latent dimension of 4 in most experiments, a choice justified empirically in Section 4.

3.5 Training Strategy

We optimize Eq. (1) with Adam (learning rate 10^{-3}), a $95/5$ train/validation split, batches of 256 frames, early stopping on the validation loss and gradient clipping. Per-instrument models are obtained under two initialization strategies. *Scratch*: each model is trained from a random initialization on its instrument only. *Checkpoint*: we first train a *base model* on all four instruments to learn a general timbral representation, and then fine-tune one model per instrument from that common starting point. Combining the two initializations with two regularization settings—AE ($\beta = 0$) and β-VAE ($\beta = 0.001$)—yields four configurations per instrument that we compare in the experiments.

3.6 Weight Interpolation

Given two models with identical architecture, a source A and a target B, and a coefficient $\alpha \in [0, 1]$, we build an intermediate model whose parameters combine those of A and B, parameter by parameter. By construction $\alpha = 0$ reproduces A and $\alpha = 1$ reproduces B, while intermediate values yield decoders that synthesize mixed timbres. We explore linear interpolation (LERP), $(1 - \alpha)\, w_A + \alpha\, w_B$, and spherical interpolation (SLERP), which traverses the arc joining both weight vectors:

$$\text{SLERP}(w_A, w_B, \alpha) = \frac{\sin\big((1 - \alpha)\theta\big)}{\sin \theta}\, w_A + \frac{\sin(\alpha\theta)}{\sin \theta}\, w_B, \tag{2}$$

where θ is the angle between w_A and w_B. When the vectors are nearly parallel, SLERP becomes numerically unstable and we fall back to LERP. Non-floating-point parameters (e.g. Batch Normalization counters) are not interpolated: we take the value of the closer model according to α.

3.7 Audio Reconstruction and Metrics

The decoder produces only a normalized magnitude spectrogram, which we rescale by X_{max} and invert to linear magnitude. Since the inverse STFT also needs phase, we evaluate Griffin–Lim, PGHI [16] (a non-iterative, single-pass phase-gradient method) and random phase, and apply the inverse STFT with the analysis window. For evaluation we use: (i) reconstruction MSE on validation, to compare latent sizes and β values; (ii) Fréchet Audio Distance (FAD) [17] over embeddings from the pretrained MERT model [18] (MERT-v1-95M), turned into a bounded similarity $\mathrm{sim} = \exp(-\mathrm{FAD}/k)$; (iii) cosine similarity between mean MERT embeddings, as an alternative baseline; and (iv) UMAP projections of the latent embeddings, with a multidimensional Wilcoxon–Mann–Whitney U statistic (0.5 = indistinguishable, 1 = fully separated) to quantify cluster separation per instrument pair.

4. EXPERIMENTS AND RESULTS

4.1 Architecture Exploration

We first fix the latent dimension. Larger latent spaces reduce reconstruction error, but for practical manipulation a small size is desirable (e.g. four control knobs are more intuitive than thirty-two). Sweeping dimensions $\{2, 3, 4, 6, 8\}$ and measuring the validation reconstruction error, a dimension of 4 gives a significant gain over 2 and 3, while 6 or 8 add little in return; the same trend holds for the β-VAE reconstruction error, for the KL divergence, and consistently across all per-instrument models and the base model. We therefore adopt latent dimension 4.

Next we select β. Sweeping $\beta \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, we examine the trade-off between the mean reconstruction MSE and the mean KL divergence on validation. Increasing regularization to $\beta = 0.01$ excessively penalizes fidelity, whereas relaxing it to 10^{-4} or 10^{-5} inflates the KL divergence toward the unregularized ($\beta = 0$) behavior without a substantial reconstruction gain. The elbow of the curve sits at $\beta = 0.001$, which we adopt; informal listening confirmed that reconstructions at $\beta = 0.001$ are very close to those at $\beta = 0$.

4.2 Latent-Space Analysis

Projecting the base-model latent embeddings to two dimensions with UMAP ($n\_neighbors = 15$) and labeling each point by instrument, the four instruments form clearly separated clusters under both $\beta = 0$ and $\beta = 0.001$, with pairwise Wilcoxon–Mann–Whitney U values close to 1. Separation is consistently slightly lower with $\beta = 0.001$, consistent with regularization producing a smoother, more continuous latent space (e.g. for the bass–voice pair the statistic drops from 0.926 at $\beta = 0$ to 0.796 at $\beta = 0.001$).

4.3 Choice of Similarity Metric

We compare cosine similarity and FAD on a voice-to-piano interpolation, using the piano reconstruction as reference and two ways of feeding the decoder: a fixed random latent Z (constant timbre) and a Z obtained by encoding a real validation clip (a trajectory through latent space). FAD stays stable and traces smooth, low-variance transition curves in both cases and across configurations, whereas cosine similarity, although it captures the general trend, is erratic along latent trajectories (spurious peaks and drops, higher dispersion). Being more robust and consistent, FAD is used for the rest of the analysis.

4.4 Weight-Interpolation Experiments

We first evaluate whether the timbre transfer holds regardless of the source and target instruments, measuring FAD similarity as a function of α for the four training configurations, averaged over many latent samples and over every source/target instrument pair. Timbre transfer occurs in all four, but their quality differs markedly. *Checkpoint-β-VAE* ($\beta = 0.001$) produces the smoothest and most sustained growth in similarity towards the target and is consistently the best configuration, followed by *Scratch-β-VAE*; *Scratch-AE* ($\beta = 0$) is the worst, failing to transition gradually and delaying the timbre change to extreme values of α. This indicates that the regularization imposed by $\beta > 0$ is the factor with the greatest impact on interpolation quality, while sharing a common weight initialization (*Checkpoint*) contributes a further improvement. Both the per-source breakdown and the aggregation over all source/target combinations reproduce this ordering, so it is not an artifact of a particular instrument pair.

A stricter test of the method is the cross-interpolation property: as the synthesized audio approaches target B it must move away from source A while *not* acquiring traits of an unrelated *witness* instrument that takes no part in the interpolation. Figure 1 reports this experiment with A = piano, B = guitar and voice as witness, tracking the FAD similarity to the three instruments as α sweeps from 0 to 1 under each configuration. The expected behaviour—similarity to A decreasing, similarity to B increasing, both curves crossing near $\alpha = 0.5$, and the witness remaining flat—is met only by *Checkpoint-β-VAE*. The *Scratch* models depart from it precisely in the intermediate range that matters for morphing: with $\beta = 0.001$ the transfer is inconsistent, and with $\beta = 0$ it is practically nonexistent until $\alpha > 0.8$, meaning the intermediate models remain locked to the source timbre instead of synthesizing hybrids. *Checkpoint-AE* does achieve the transfer, but abruptly and with erratic fluctuations in the witness curve, indicating that its intermediate decoders drift towards timbres unrelated to either endpoint. These results confirm that a common initialization combined with a regularized latent space

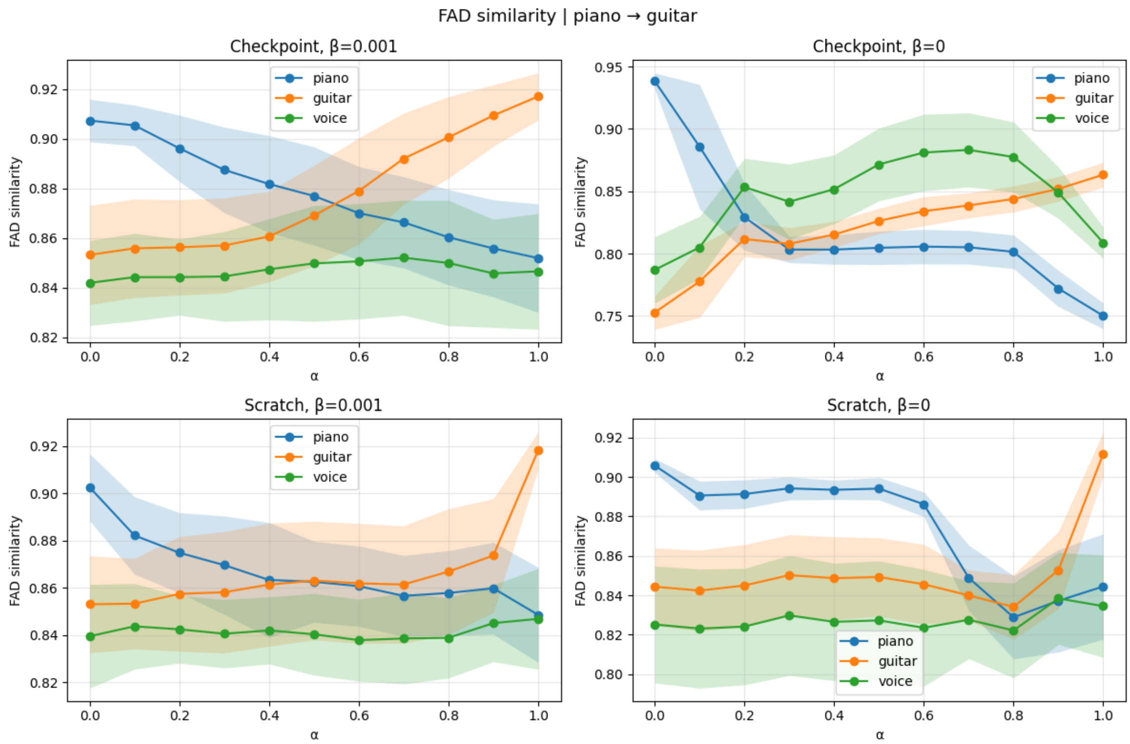

Figure 1: Cross interpolation from piano (source A) to guitar (target B) with voice as a witness instrument, for the four training configurations: FAD similarity to each of the three instruments as the interpolation coefficient α sweeps from 0 to 1. Only *Checkpoint-β-VAE* (top left) exhibits the expected behaviour—piano decreasing, guitar increasing, a crossover near $\alpha = 0.5$ and a flat witness. *Checkpoint-AE* (top right) transfers abruptly and perturbs the witness, while the *Scratch* models (bottom) barely change until α approaches 1.

(*Checkpoint* with $\beta = 0.001$) is what yields a predictable, gradual and independent timbre transfer.

4.5 Audio Examples

To complement the quantitative analysis with a perceptual assessment, audio examples of continuous timbre transitions (voice-to-piano and guitar-to-piano, using the *Checkpoint-β-VAE* configuration) and of audio-to-audio morphing between real recordings are provided as supplementary material.

5. CONCLUSIONS

We developed and validated a technique to synthesize intermediate timbres between instruments by interpolating the weights of specialized VAE models. Unlike conventional techniques that operate within the latent space of a single model trained on multiple instruments, our approach trains an independent model per instrument and interpolates directly over its parameterization, producing intermediate decoders that generate hybrid timbres. A latent dimension of 4 offers the best trade-off between reconstruction fidelity and manipulability, and $\beta = 0.001$ sits at the elbow of the reconstruction–KL curve. The central finding is that interpolation quality depends strongly on the training configuration: regularizing the latent space (β-VAE) and starting from a common base model via fine-tuning (*Checkpoint-β-VAE*) consistently produced the smoothest and most predictable transitions, both in the FAD metric and in subjective listening. FAD over MERT embeddings proved a robust similarity metric, outperforming cosine similarity in stability.

A deliberate design choice was to work with small models and a limited amount of data (about 87 minutes per instrument, four instruments), so as to keep the models easy to train, lightweight and runnable on modest hardware—aligned with the goal of a future real-time synthesis tool. As a counterpart, this limits the attainable fidelity and generality; we view it as a conscious balance between quality and lightweight, real-time viability rather than an intrinsic limitation of the method. Future work includes real-time inference and weight interpolation, integration into audio-production tools (e.g. VST plugins), enlarging the dataset and instrument set, and extending the technique to a triple interpolation controlled by two parameters.

6. ETHICS STATEMENT

This work uses publicly available datasets (GuitarSet, FiloBass, VocalSet) under their respective licenses, together with public-domain piano recordings, and does not involve the collection of personal or private data. The proposed generative technique is intended as a creative tool for music production and performance; as with any audio-synthesis method, care should be taken to respect the rights of performers whose recordings are used to train the models.

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